Health

Smartphone sensors: challenges and opportunities

Health Research from Home, 2026

Matthew Thompson
Clinical Research Scientist, Google
Emeritus Professor of Family Medicine, University of Washington, Seattle

Potential health sensing from three sensors found in many smartphones:



Camera

Non-contact health monitoring, imaging analysis.



Microphone

Audio biomarkers, cough & respiratory analysis.



Ultrawideband Radar

Motion tracking, vital sign detection, heart rate.

Opportunities and challenges from industry/health research perspective

Conflicts of interest: Own Alphabet stock

1) Smartphone camera

Smartphone images have been extensively researched for consumer and clinical value.

Some have entered widespread use:

- Skin lesion identification
- Retinopathy screening

Question: what would it take to detect **anemia** using smartphones?

Anemia (low hemoglobin or Hb)

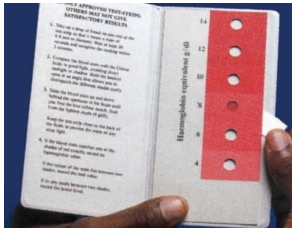
Global

- > 50% of children <5 yr and pregnant women are anemic in some countries
- Targeted interventions can resolve the majority (iron, helminths, malaria etc)
- Only 3 of 82 LMICS were on target to reach WHO goals of reducing anemia by half by 2025

Higher income countries

- Hb one of the most common blood tests done by GPs
- Fatigue, cancer detection, disease monitoring, pregnancy









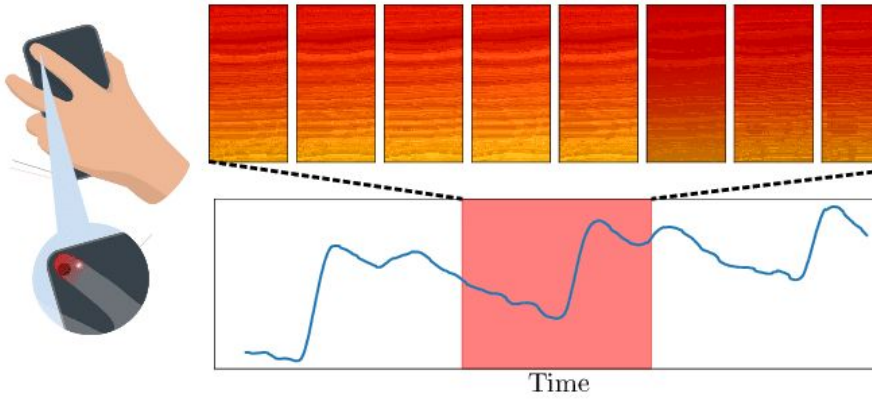
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Potential to detect anemia/estimate Hb from smartphone sensors

Detect pallor from **camera images** of parts of the body (nailbed, tongue, conjunctiva)



Photoplethysmogram (PPG) signals measure reflectance or transmittance of LED light signals from blood in the fingernail bed



Seeing Red: PPG biometrics using smartphone cameras. Lovisotto G et al, April 2020

Data collection for smartphone anemia detection (funded by Gates Foundation)

Screening for anemia using multi-modal machine learning models on smartphones: protocol for a comparative accuracy study in rural India

Shrey Desai, Kaushal Jesalpura, Sujay Kakarmath, Mayank Daswani, Sethuraman Venkatraman, Jim Taylor, Shruti Gadgil, Marc Wilson, Rajroshan Sawhney, Anaita Singh, Raghu Pullakhandam, Matthew Thompson

doi: https://doi.org/10.1101/2025.04.10.25325591

Setting: SEWA Rural hospital and outpatient clinics, serving a rural, tribal population in Gujarat, India.

Population:

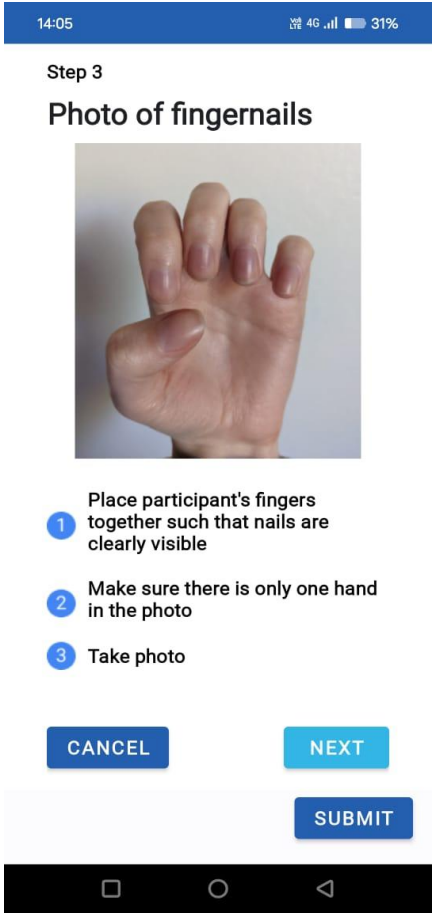
- Adults and children (**age 1-89 years**) for whom a Complete Blood Count was ordered during routine care.
- Recruitment targeted specific hemoglobin (Hb) levels (normal, **mild, moderate, severe anemia**)

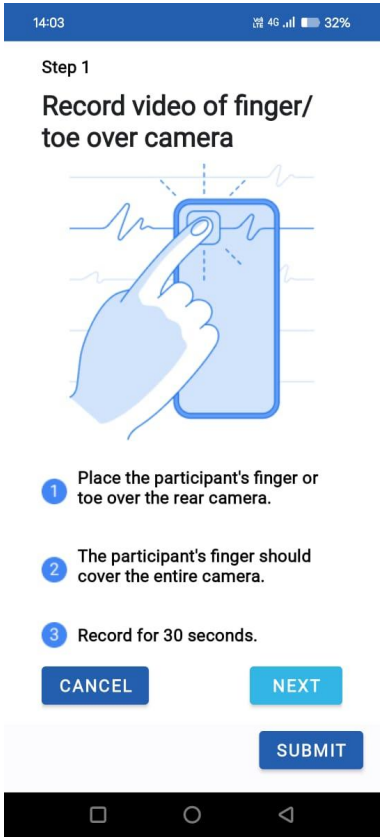
Smartphone-Based Data Collection

- Smartphone images of **conjunctiva**, **tongue**, and **fingernails** using **digital data collection tool.**
- 30-second photoplethysmogram (**PPG) video** recording from the index finger.
- Data collected using **three** different smartphone models (Samsung A14, A05, S24)

Reference/Comparator Tests:

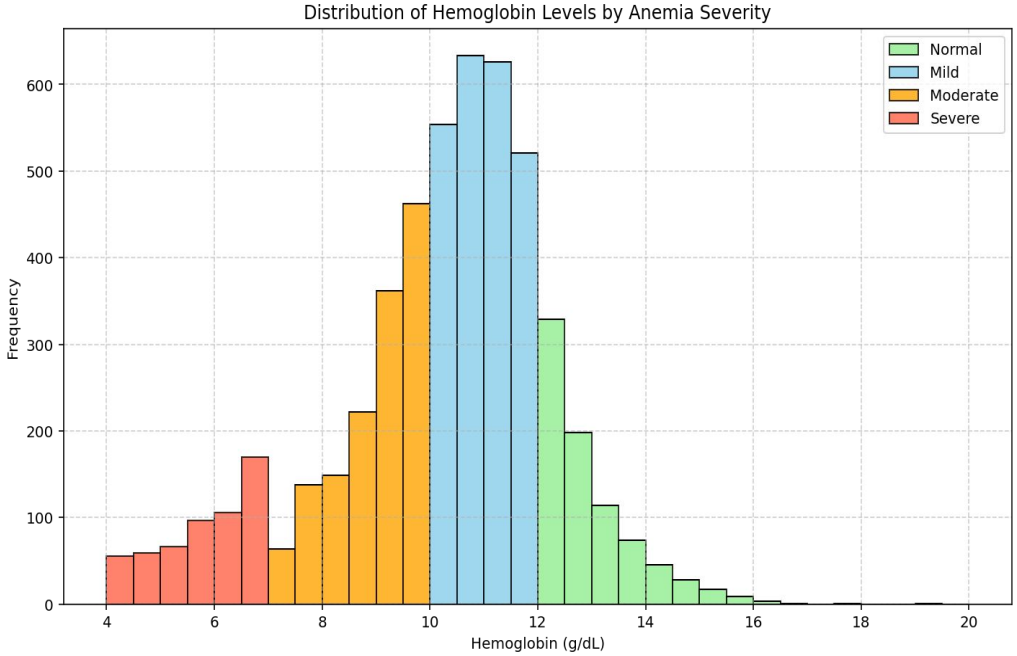
- **Primary Reference**: Auto-analyzer (Sysmex XN-550) with strict quality control.
- **Comparators**: Point-of-care Hb test (HemoCue Hb 301)


14:05
31%
Step 3
Photo of fingernails
1 Place participant's fingers together such that nails are clearly visible
2 Make sure there is only one hand in the photo
3 Take photo
CANCEL
NEXT
SUBMIT


14:03
32%
Step 1
Record video of finger/ toe over camera
1 Place the participant's finger or toe over the rear camera.
2 The participant's finger should cover the entire camera.
3 Record for 30 seconds.
CANCEL
NEXT
SUBMIT

Health

Data Characteristics



Results of accuracy

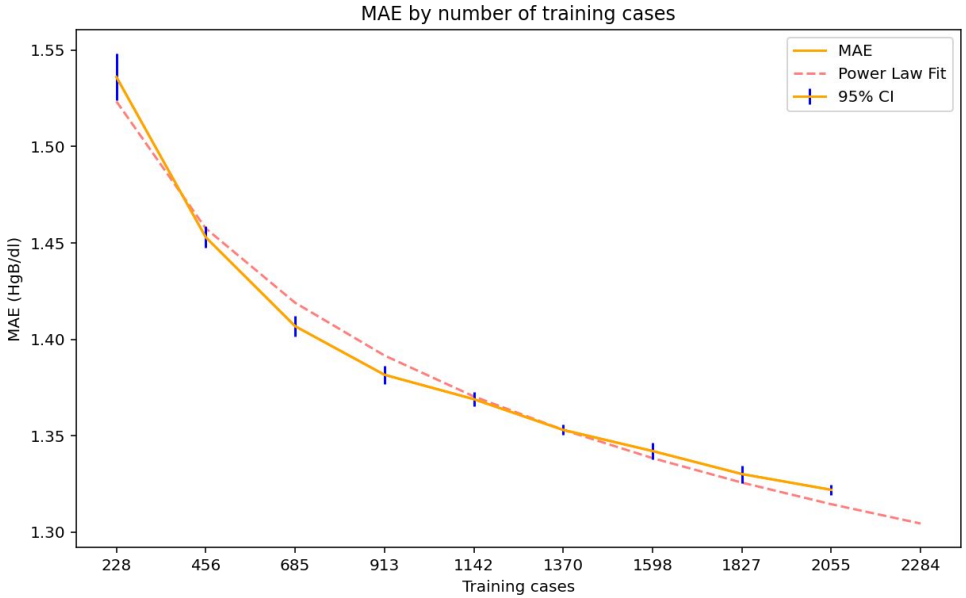
Anemia vs No Anemia	**Sensitivity**	74%
	Specificity	70%
Severe/Moderate vs Mild/Normal	**Sensitivity**	81%
	Specificity	71%
Severe Anemia vs Mild/Moderate/Normal	**Sensitivity**	96%
	Specificity	70%

Note: Classification results are secondary calculations from a regression model.

How close to achieving blood test level accuracy?

Takeaways

- MAE improvement with additional data follows a power law distribution.
  - Estimated Mean Absolute Error:
    - n=1000: 1.38 g/dL
    - n=2284: 1.30 g/dL (actual: 1.29)
    - n=5000: 1.23 g/dL
    - n=10,000: 1.16 g/dL



Smartphone camera for anemia detection

Data size

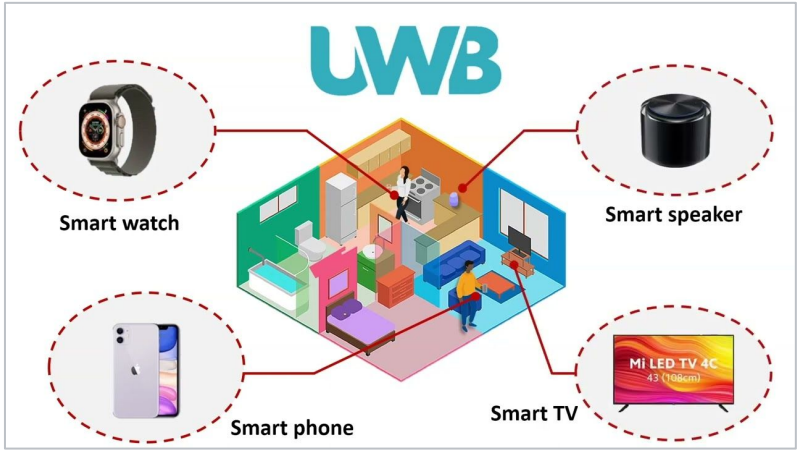
- Huge data sets needed, rigorously obtained
- Diminishing returns of data volume vs model accuracy

Commercialization

- Good enough? For what setting / purpose?
- Value- varies with settings, unlikely to meet laboratory level accuracy
- Motivations and burden of commercializing

2) Ultra wide band (UWB) radar

- UWB technology present in range of consumer electronics
- Many smartphones have multiple radar emitters and receivers
- Used for localization and tracking (including devices such as AirTags or SmartTags), vehicle unlock features, or data transfer.



RESEARCH-ARTICLE | OPEN ACCESS

Embracing Consumer-level UWB-equipped Devices for Fine-grained Wireless Sensing

Authors: Fusang Zhang, Zhaoxin Chang, Jie Xiong, Junqi Ma, Jiazhi Ni, Wenbo Zhang, Beihong Jin, Daqing Zhang | Authors Info & Claims

Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies, Volume 6, Issue 4
Article No.: 194, Pages 1 - 27 • https://doi.org/10.1145/3569487

Question: can UWB Radar accurately measure heart rate?

Key value (for GPs, Nurse triage, parents of children, individuals)

- Without contact
- Through clothing
- Irrespective of skin pigmentation

Chest wall movements are easy observed.

What about the **micro-movements from heart/arterial pulsations**?

Can we differentiate heart/arterial pulsations from breathing movements and environmental noise?

UWB RADAR-BASED HEART RATE MONITORING: A TRANSFER LEARNING APPROACH

Elzbieta Gruzewska[1,†], Pooja Rao[1], Sebastien Baur[1], Matthew Baugh[1,2,*], Mathias M.J. Bellaiche[1], Sharanya Srinivas[1], Octavio Ponce[1], Matthew Thompson[1], Pramod Rudrapatna[1], Michael A. Sanchez[1], Lawrence Z. Cai[1], Timothy JA Chico[3,4], Robert F. Storey[3,4], Emily Maz[3,4], Umesh Telang[1], Shravya Shetty[1], Mayank Daswani[1]

[1]*Google*
[2]*Imperial College London*
[3]*NIHR Sheffield Biomedical Research Centre, Sheffield Teaching Hospitals NHS Foundation Trust, Sheffield, United Kingdom*
[4]*Division of Clinical Medicine, University of Sheffield, Sheffield, United Kingdom*
[*]*Work performed while interning at Google Research*
[†]*Corresponding author: epawelczyk@google.com*

Data from Sheffield, Savannah (GA), Mobile (AL)

- 376 participants at rest
- NXP SR160 UWB radar device
- HR from ECG lead 1 or PPG signals form Biosignals Plux system
- 2 min recordings from 3 positions



a) Radar on the table in front of the participant.

b) Radar on the participant's lap.

c) Radar on the table

Figure 1: Three radar positions used during the IR-UWB

Table 11: Model performance on IR-UWB radar data including 95% confidence interval by gender.

Gender	Test MAE [95% CI]	% of test samples
Overall	4.1 [3.8, 4.5]	–
Female	4.4 [3.9, 4.9]	74
Male	3.5 [2.9, 4.1]	26

Accuracy

- Potentially adequate, with more research on faster heart rates, different age groups, confounders

Competing priorities

- Is there sufficient value for smartphone manufacturers, given other ways of measuring heart rate in adults (smartwatches, earbuds, rings)?

Utility

- Is there sufficient utility for clinical users, and for home use?

3) Microphone - cough detection

- **Common symptom** prompting health care visits
- Acute cough– upper resp infections
  - Adult 1-2 colds per year, child 6-8 colds per year
- Chronic cough
  - Chronic lung disease (asthma, COPD), heart disease (heart failure) monitoring
  - Smoking, lung cancer, medication side effects
- Subjective assessment (cough scores etc) inaccurate

- **Pharma trials of drugs for which cough noted as side effect**
  - > 70 pharma trials - medication side effects, asthma, COPD, etc
- **Lung cancer**
  - Better survival with some new types of drugs (EGFR inhibitors), key sign of relapse is increasing cough
  - >2,000 current lung cancer trials in US

What does a smartphone cough detection tool need?

Proprietary + Confidential

- Can cough sounds be **differentiated** from other audio signals?
- How accurately can the sound of a cough be **attributed** to one person?
- Under which **user conditions and/or at what times** in a 24 hour period can cough be identified most accurately and efficiently?
- **Privacy issues and acceptability** of monitoring sound?
- Demonstrating utility ie is it '**better**' than current practice (***better = more actionable, more acceptable, more useful, improves health outcomes etc***)

Smartphone-based cough detection predicts asthma control – description of a novel, scalable digital biomarker

Peter Tinschert | Frank Rassouli | Filipe Barata Show More

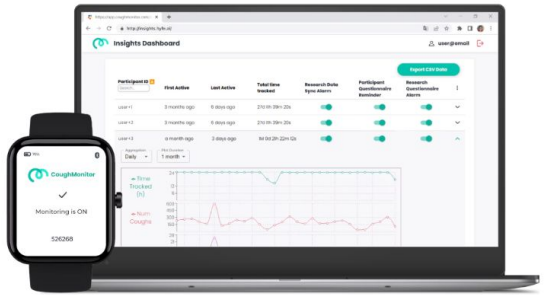
European Respiratory Journal 2020 56(suppl 64): 4569; DOI: https://doi.org/10.1183/13993003.congress-2020.4569

Hyfe Solutions Products Resources Indications Company Schedule A Call

Integrate Cough as an Endpoint with CoughMonitor Suite

The CoughMonitor Suite is a comprehensive research tool for measuring cough as an endpoint in clinical trials. A slim, lightweight watch collects reliable, validated, continuous cough data while patients go about their normal daily lives.

Talk To A Cough Monitoring Expert



Smartphone-Enabled Cough Detection in Severely Exacerbated COPD: An Exploratory Pilot Study

https://doi.org/10.6084/m9.figshare.31113083

Cite Download all (1.32 MB) Share Embed + Collect

Research

Diagnosing community-acquired pneumonia via a smartphone-based algorithm: a prospective cohort study in primary and acute-care consultations

Paul Porter, Joanna Brisbane, Udantha Abeyratne, Natasha Bear, Javan Wood, Vesa Peltonen, Phillip Della, Claire Smith and Scott Claxton

British Journal of General Practice 2021; 71 (705): e258-e265. **DOI:** https://doi.org/10.3399/BJGP.2020.0750

Aim: To develop and test a **smartphone-based algorithm for diagnosing CAP** without need for clinical examination or radiological inputs.

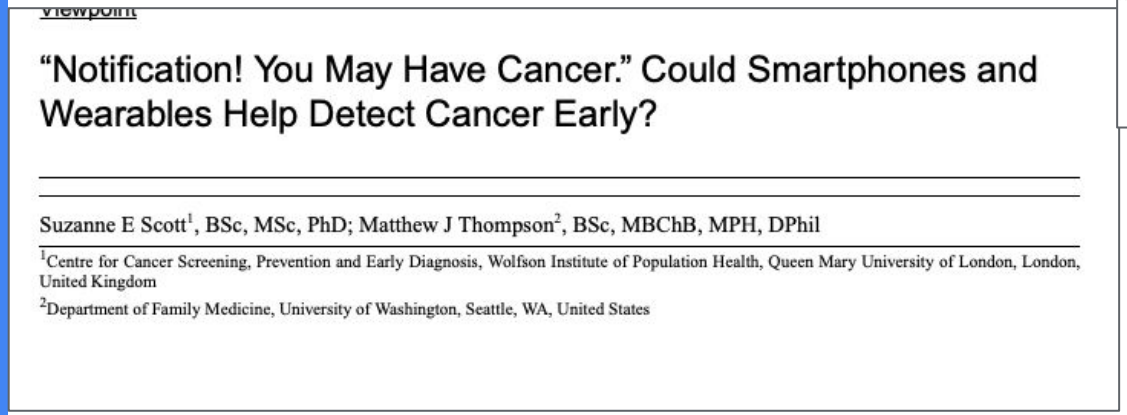
Design and setting: A prospective cohort study using data from participants aged >12 years presenting with acute respiratory symptoms to a hospital in Western Australia.

Method: Five **cough audio-segments were recorded and four patient-reported symptoms (fever, acute cough, productive cough, and age)** were analysed by the smartphone-based algorithm to generate an immediate diagnostic output for CAP.

Results: The smartphone-based algorithm had high percentage agreement (PA) with the clinical diagnosis of CAP in the total cohort (*n* = 322, positive PA [PPA] = 86.2%, negative PA [NPA] = 86.5%, area under the receiver operating characteristic curve **[AUC] = 0.95**

Opportunities and challenges: summary points

How close / far are we from smartphone/wearabke tools that will tackle the most important healthcare problems in the UK?



Viewpoint

"Notification! You May Have Cancer." Could Smartphones and Wearables Help Detect Cancer Early?

Suzanne E Scott[1], BSc, MSc, PhD; Matthew J Thompson[2], BSc, MBChB, MPH, DPhil

[1]Centre for Cancer Screening, Prevention and Early Diagnosis, Wolfson Institute of Population Health, Queen Mary University of London, London, United Kingdom

[2]Department of Family Medicine, University of Washington, Seattle, WA, United States

Health and Social Care Secretary, Wes Streeting, said:

"Cancer survival shouldn't come down to who won the lottery of life. But cancer is more likely to be a death sentence in Britain than other countries around the world.

Potential signals of cancer from smartphones or wearables

Table 1.

Potential signals of cancer that can be measured using sensors in smartphones or wearables.

Sensors currently available on some smartphones or wearables	Examples of signals for health features that could be related to cancer
Audio signals from microphones [13]	• Changes in cough and breathing difficulty (associated with lung cancer) [14] • Changes in voice such as hoarseness (associated with head and neck cancer and lung cancer) [14]
GPS location tracking and activity tracking [15]	• Reduced activity resulting from fatigue (associated with multiple cancers) [14]
Image capture and analysis [16-18]	• Anemia detected from images of the skin or eyes (associated with multiple cancers) [19] • Jaundice detected from images of the skin or eyes (associated with pancreatic cancer) [20] • Changes in skin lesions (associated with skin cancer) [14]
Temperature measurement [21]	• Rise in temperature (associated with pancreatic cancer) [22]
Body composition using image analysis and electro dermal activity [23]	• Weight loss (associated with multiple cancers) [14]
Photoplethysmogram [24]	• Anemia (associated with multiple cancers) [19]

Google

BUT.....what would it take?

Signals and sensors with accuracy & utility	• Selecting signals, developing sensors, generating evidence of accuracy and utility • Competing commercial priorities
Regulatory	• Evolving regulatory environment. • Safety, quality, acceptability, credibility, real world performance.
Equity	• Views and needs of communities • Biases based on skin tone or other differences. • Availability and quality across brands and models.
Acceptability & adherence	• User burden, technical issues, drop out, follow up . • Cost motivation, comfort, ease of use, trust in data, interpretability • Willingness to share data with researchers, doctors, companies.
Unintended consequences	• False positives- over investigation, costs, anxiety. • False negatives - false reassurance

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